Computers and Electronics in Agriculture 225 (2024) 109238

Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag

Coupled WOFOST and SCOPE model for remote sensing-based crop growth simulations

Georgios Ntakos [a,b,*], Egor Prikaziuk [a], Tamara ten Den [c,e], Pytrik Reidsma [c], Nastassia Vilfan [d], Tamme van der Wal [e,f], Christiaan van der Tol [a]

[a] Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, P.O. Box 217, 7500 AE, Enschede, The Netherlands
[b] Business Unit Greenhouse Horticulture, Wageningen University & Research, P.O. Box 644, 6700 AP, Wageningen, The Netherlands
[c] Plant Production Systems, Department of Plant Sciences, Wageningen University & Research, P.O. Box 430, 6700 AK, Wageningen, The Netherlands
[d] Horticulture and Product Physiology, Department of Plant Sciences, Wageningen University & Research, P.O. Box 16, 6700 AA, Wageningen, The Netherlands
[e] Agrosystems Research, Wageningen University & Research, P.O. Box 430, 6700 AA, Wageningen, The Netherlands
[f] AeroVision B.V., 3811 HN, Amersfoort, The Netherlands

ARTICLE INFO

Keywords:
SCOPE model
WOFOST model
Remote sensing data assimilation
Hyperspectral reflectance
Crop growth simulation

ABSTRACT

Crop yield prediction plays an important role in food security. Crop growth models can be used for this purpose, however, they require empirical parametrization. In this study, we show that remote sensing observations can be used to parameterize a crop growth model, providing crop growth monitoring and yield predictions, while mitigating the need for extensive field measurements and accounting for variability in crop varieties. For this purpose, we coupled the SCOPE radiative-transfer model with the state-rate crop model WOFOST. In this paper, we explain how the two models are coupled, assess the accuracy and present a global sensitivity analysis of the coupled model. To assess the accuracy of the coupled model in predicting growth and yield, experimental data of five potato varieties from two fields, in two different locations, in the Netherlands were used. Results show that simulated LAI had a good correlation with measured LAI for four out of the five varieties ($R^2 > 0.6$), while RMSE was in the range 1.09–1.34 m^2 m^{-2}. Simulated leaf dry matter content showed similar results, with high accuracy for all varieties ($R^2 > 0.6$) and RMSE between 310 and 440 kg ha^{-1}. Finally, yield simulation showed promising results with an average difference of 1800 kg ha^{-1} between simulated and measured values. Based on the findings of this study, it is possible to conclude that a coupled WOFOST-SCOPE model has the potential to improve the application of remote sensing data for crop growth monitoring and yield prediction.

1. Introduction

Over recent decades, technological advancements have significantly improved agricultural monitoring and management capabilities. Despite these advancements, the agricultural sector faces pressing challenges such as climate change impacts, soil degradation, and the increasing demands of a growing global population (Radoglou-Grammatikis et al., 2020). Moreover, even though available tools perform relatively well at regional levels, at field level, the prediction accuracy remains limited (Silva and Giller, 2020). To address these challenges, there is a pressing need for advanced digital tools that can facilitate informed decision-making in agriculture.

Two primary tools, crop growth models and remote sensing, have been extensively utilized in the agricultural domain. Crop growth models offer mechanistic frameworks for simulating plant growth processes and predicting yields based on various input parameters. On the other hand, remote sensing serves as an instrumental method for vegetation monitoring, providing spatial and temporal data on vegetation health and dynamics. Historically, these tools have been utilized both independently and in combination, each offering unique advantages and applications. While the assimilation of remote sensing variables into crop growth models has a long history, the relatively recent increase in the availability of satellite data with high spatio-temporal resolution is creating new opportunities. By coupling these enhanced remote sensing datasets with crop growth models, it is possible to scale predictions and assessments to the field or within-field level, thereby greatly improving utility (Dlamini et al., 2023).

* Corresponding author at: Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, P.O. Box 217, 7500 AE, Enschede, The Netherlands.

E-mail addresses: g.ntakos@utwente.nl, georgios.ntakos@wur.nl (G. Ntakos), e.prikaziuk@utwente.nl (E. Prikaziuk), tamara.tenden@wur.nl (T. ten Den), pytrik.reidsma@wur.nl (P. Reidsma), nastassia.vilfan@wur.nl (N. Vilfan), tamme.vanderwal@wur.nl (T. van der Wal), c.vandertol@utwente.nl (C. van der Tol).

https://doi.org/10.1016/j.compag.2024.109238
Received 29 February 2024; Received in revised form 9 June 2024; Accepted 7 July 2024
Available online 20 July 2024
0168-1699/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

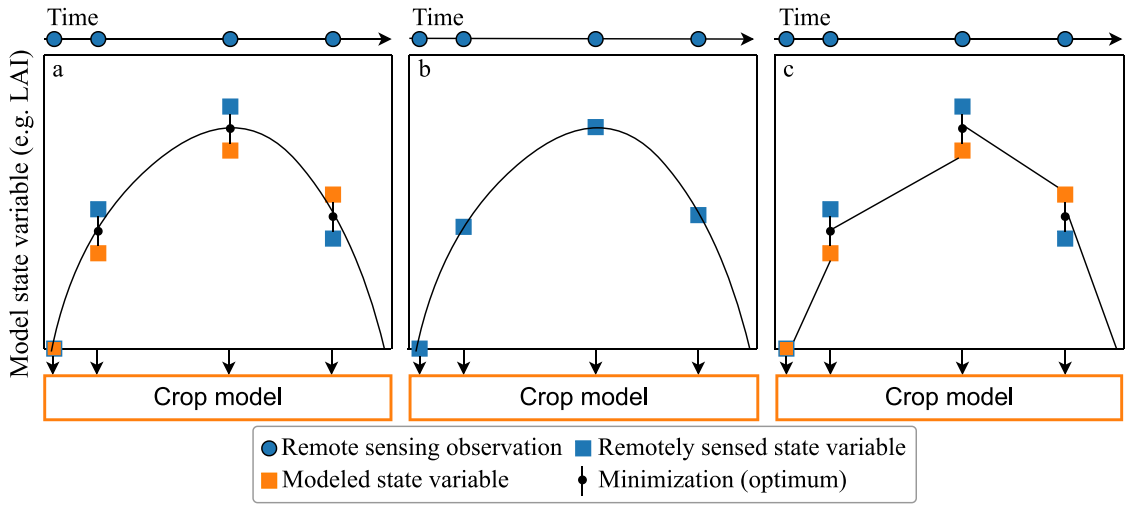

Fig. 1. Representation of remote sensing data assimilation (**a**) Model calibration (or re-calibration), (**b**) Forcing method and (**c**) updating method (Image acquired from Jin et al. (2018)).

Crop growth models, such as the WOrld FOod STudies model (WOFOST) (Boogaard et al., 2014), have emerged as tools for forecasting physiological development, growth, and yield (Jin et al., 2018). These point-scale, state-rate crop growth models can leverage data on weather, soil, crops and management to simulate crop growth and yield (Siad et al., 2019). They are applied across various agricultural production domains (Holzworth et al., 2015), such climate change impact (Faye et al., 2023) and adaptation (Reidsma et al., 2015), decision support (Nicoleta Tantalaki and Roumeliotis, 2019) and breeding (Cooper and Messina, 2022).

Crop growth models are designed to assess the environmental effects on plant physiological processes, enabling precise simulations of crop growth and development in predetermined intervals, typically daily. This leads to an accurate estimation of crop yield at the time of harvest. Prior to deploying such models, a localized calibration is essential, usually based on in situ observations, literature reviews, or expert insights (Ceglar et al., 2019; Van Ittersum et al., 2013). Generally, crop modelers calibrate an average of six genotype-specific growth parameters, regardless of the model, rendering the models most effective under conditions akin to the calibration settings (Seidel et al., 2018). There are no clear guidelines on this process but, the overwhelming majority of studies on calibrating crop models, based on remotely sensed state variables, focus on parameters connected to LAI, sowing date and above-ground biomass (Dlamini et al., 2023). However, the crop growth simulation accuracy diminishes when the models are being applied to dissimilar varieties or regions (Akhavizadegan et al., 2021). Moreover, the accuracy of the crop models is reduced when applied to heterogeneous conditions and at higher resolutions, due to them being developed mostly for applications on field scales with homogeneous conditions (Ceglar et al., 2019).

Remote sensing facilitates efficient and precise agricultural monitoring across extensive regions with consistent revisits, positioning it as a potential alternative to manual calibration (Atzberger, 2013). European Space Agency's (E.S.A.) Sentinel 2 (Drusch et al., 2012), an open, multispectral, fine resolution satellite sensor, has been extensively utilized for crop monitoring (Kaplan et al., 2023; Qader et al., 2023; Shen et al., 2023; Li et al., 2022). The German Environmental Mapping and Analysis Program (EnMAP) (Guanter et al., 2015) has the potential to provide the scientific community with high-quality hyperspectral satellite images. Furthermore, the imminent FLEX/Sentinel-3 TandEM mission promises to revolutionize the domain of vegetation monitoring by offering unparalleled insights into vegetation dynamics, leveraging sun-induced fluorescence and reflectance (E.S.A., 2024). However, since satellite-based sensors can cover a large range in spatial resolution, with up to 300 m for FLEX, not all options meet the requirements for field-based monitoring.

With the introduction of radiative transfer models, such as PROSAIL (Jacquemoud et al., 2009), researchers can utilize remotely sensed spectral reflectance measurements to extract critical biophysical parameters at both foliar and canopy scales (Jacquemoud et al., 2009). By combining vegetation radiative transfer and crop models, remote sensing data can be used as an additional constraint to further refine crop model predictions. Aligning with this approach, researchers have integrated remote sensing data through data assimilation designs into crop models for years, underscoring the synergistic benefits of these combined methodologies (Kasampalis et al., 2018). In this direction, accuracy of crop growth simulations is a metric, influenced by the spatial and temporal resolution of the remote sensing observations. Curnel et al. (2011) shows that, yield simulation accuracy increases when the temporal resolution of the observation is high, between three and seven days, compared to bigger time steps of 15 and 30 days. Similarly, in Huang et al. (2019b), a small but clear increase of accuracy is observed when using Landsat (30 m) compared to MODIS (1 km). Furthermore, Curnel et al. (2011) argues that, smaller error in the observation is much more important than a higher resolution and that the assimilation method can significantly influence the crop growth simulation.

For the assimilation of data into crop models, three predominant methodologies have been formulated and assessed over the years: calibration, forcing and updating (Jin et al., 2018) (Fig. 1). The calibration method seeks to minimize the difference between model simulations and state parameter observations, typically by re-initializing the model with varied parameter values using an optimization algorithm (Huang et al., 2019a). The forcing method replaces simulated state variables with values derived from remote sensing observations, bypassing the crop growth model's simulated data (Tripathy et al., 2013). The updating method adjusts the model's state variables with each new observation, with the crop model calculating state parameter values between two observations. This method assumes that enhanced data on a given day can improve subsequent days' accuracy. It is favored for its computational efficiency and reliable results, though observation gaps can impact accuracy (Jin et al., 2018).

In data assimilation strategies, various biophysical variables have been used, with the overwhelming majority of the studies focusing on Leaf Area Index (LAI) while other variables include fraction of absorbed photosynthetically active radiation (FAPAR), soil moisture and evapotranspiration, either individually or in combination (Dlamini

et al., 2023). In data assimilation, certain variables, such as LAI, can be directly integrated into crop growth models. However, other variables, like leaf dry weight, need to be converted before assimilation while in some instances, there is a mismatch between the retrievable data and the requirements of the crop models, as seen with chlorophyll content. The values of the assimilated variables can be retrieved through manual measurements or by inversion of radiative transfer models, though the latter necessitates high accuracy in remotely sensed trait estimation (De Wit et al., 2012).

Alternatively, an underexplored approach involves the integration of crop growth models with radiative transfer models, permitting the direct assimilation of reflectance measurements and circumventing the need for trait retrieval. While a lot of studies focus on the assimilation of vegetation indices (Launay and Guerif, 2005; Ma et al., 2013b), only a handfull, such as the study of Huang et al. (2019b), investigate the assimilation of reflectance. While methodologies involving vegetation indices show impressive results, and on top of it rely on widely accessible multispectral data, hyperspectral remote sensing is booming and with computational power being less of a problem, the extra spectral information, combined with RTMs, can even outperform vegetation indices, if used properly (Kasampalis et al., 2018).

This study aims to present the coupling of the WOFOST crop growth model with the SCOPE radiative transfer model. By doing so, we seek to enhance the simulation of crop growth and improve the retrieval of plant traits. The WOFOST model simulates time series of biophysical variables, which SCOPE uses to generate the corresponding spectral reflectance. Moreover we want to present our methodology for optimizing the coupled WOFOST-SCOPE model's input parameters using remote sensing data assimilation techniques, allowing for refined simulations.

Our approach is distinct in several ways. We go beyond the typical reliance on leaf area index (LAI) by incorporating a variety of biophysical variables, offering a more comprehensive simulation. Furthermore, this is the first study to couple WOFOST with the SCOPE RTMo (FLUSPECT + SAIL) framework, for mechanistic crop growth simulations, based on spectral reflectance. Additionally, our methodology is the first one, to the extend of our knowledge, that applies a coupled radiative transfer and crop model to potato crops, that have the peculiarity of a significant variable, the tuber growth, not being visible from the top of canopy.

First, we introduce and describe our integrated modeling approach. Following this, we proceed to investigate whether incorporating top-of-canopy reflectance data, obtained via remote sensing, enhances the model's ability to accurately simulate key crop attributes such as Leaf Area Index (LAI), dry matter content, and overall yield. To conduct this evaluation, we utilize spectral reflectance measurements gathered from five different varieties of potatoes, cultivated in two regions in the Netherlands. Finally, for validation purposes, we compare the coupled model's output with data obtained from destructive field measurements.

2. Materials and methods

2.1. Study area

The data for this study was collected during the summer of 2019 in two locations in the Netherlands, namely Lelystad (52.54 °N, 5.55 °E) and Vredepeel (51.54 °N, 5.86 °E) (Fig. 2). Both fields belong to an experimental farm of Wageningen University and Research, located in regions commonly used for potato cultivation by commercial farmers. They serve as typical examples of arable production sites in their respective regions and are characterized by homogeneous soil conditions: light clay soil in the Lelystad field and sandy soil in the Vredepeel field.

In both experimental fields, plots with size of 9 m × 12 m or 15 m^2 for Lelystad and 6 m × 21 m or13.5 m^2 for Vredepeel were designed, uniformly, featuring five distinct potato varieties. In each field, all the varieties were planted simultaneously, regardless of their cultivation duration. For this study, data was exclusively sourced from plots with optimal irrigation and fertilization, focusing on potential yield (Table 1).

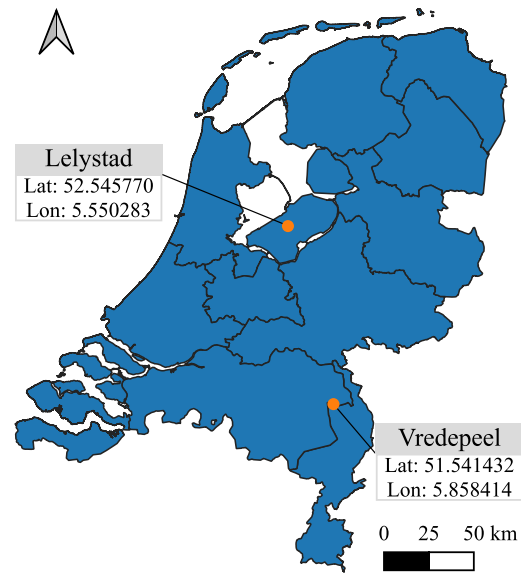

Fig. 2. The location and coordinates of the two fields used for this study.

Table 1
Information about the plots and fields, used in this study.

Location	Cultivars	Plot numbers	Nitrogen (kg/ha)	T. Irrigation (l/ha)
Lelystad	Innovator	02, 25, 42	375	1643.5
	Fontane L	07, 22, 38	360	1643.5
	Markies	05, 28, 36	320	1643.5
Vredepeel	Premiere	13, 30, 40	320	1191.3
	Fontane V	01, 10, 18	320	1191.3
	Festien	24, 25, 34	265	1191.3

2.2. Experimental data

During the potato growing season, field measurements were conducted in the two experimental fields, from June to September 2019. The cultivated varieties were Fontane (referred to as Fontane L), Innovator and Markies, in Lelystad, and Fontane (referred to as Fontane V), Premiere and Festien in Vredepeel. Three repetition plots were used for each variety in each location.

Detailed information about the agro-management practices (fertilization, irrigation), emergence, tuber initiation, maturity, harvest days and final yield were logged. The weather data were obtained from weather stations close to the experimental fields, while soil characteristics were also measured.

In-situ measurements were obtained every two weeks until each variety reached the mature stage. From 4 plots (2 plots for Fontane and 1 plot for the rest two varieties per field), a compound leaf was taken and lab-based, destructive measurements, were conducted. In the lab, three extractions of leaf chlorophyll content (Cab) and leaf carotenoid content (Cca) per leaflet were conducted. Moreover, Leaflet area (A) and fresh weight (FW) and was measured and then were dried for 48–72 h at 60 degrees Celsius, to measure the their dry weight (DW) as a ratio of dry leaf mass to leaf area (Cdm). Leaf water content (Cw) was computed based on the equation: $Cw = (FW - DW)/A$. The protocols for these measurements are being described in Prikaziuk et al. (2022). In the same experiment, another set of destructive measurements was conducted and described in ten Den et al. (2022), including LAI and the dry weight of living leaves (WLV) and stems (WST). These measurements were conducted on plants harvested from an area of around 3 m^2, per measurement day, per plot, leaving approximately 13.5–15 m^2 for the final yield harvest.

Within the experimental field, relative chlorophyll leaf measurements (PS) were obtained with the MultispecQ device by PhotosynQ

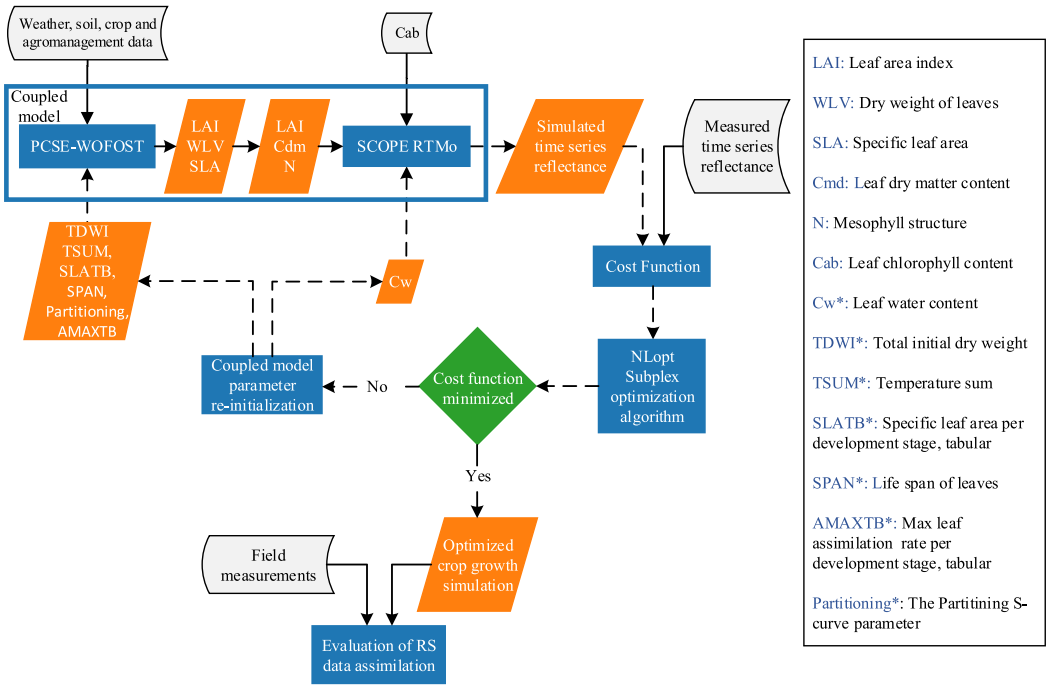

Fig. 3. Flowchart of the couple WOFOST – SCOPE RTMo model data assimilation framework. The white color refer to input data, the blue color refers to a process, the orange color refers to process output while the green refers to a condition loop. The legend describes the included acronyms and the parameters with asterisk (*) are the parameters that the optimization can re-initialize.

(2024) from five random, fully sunlit leaves per plot. These measurements were used, combined with the destructive chlorophyll content measurements, to estimate the leaf chlorophyll content for each variety and plot, based on a calibrated curve ($Cab = 0.053 \cdot PS^{1.67}$). This curve aligned well with other reported literature-based relationships, as mentioned in Prikaziuk et al. (2022). Finally, two LAI measurements were obtained for each plot with the LAI-2000 Plant Canopy Analyze (Li-Cor, 2024), following the standard protocol written in the device manual.

The Hyperspectral measurements were carried out using the A.S.D. FieldSpec 4 device (Panalytical, 2023), positioned at the top of the canopy. The device was fixed 1.5 m above the ground, and five radiance measurements were taken per plot on each measurement day. These measurements were acquired in succession, parallel to the cultivated rows, following the longer direction of each plot. The measurements began approximately half a meter within the plot, excluding the first two rows and any empty areas resulting from intermediate harvests, to avoid border effects. For the Hyperspectral measurements the same protocol was followed were, the resulting radiance for each measurement was the average of 10 consecutive measurements, while the white reflectance panel was measured before and after each canopy measurement and linearly interpolated to the time of each canopy measurement. The device was re-calibrated for varying light conditions at regular intervals. The reflectance was then calculated from the raw A.S.D. digital number measurements. The reflectance produced covered the range from 400 nm up to 2500 nm, with a spectral resolution of 1 nm and the atmospheric windows were removed (1350–1430 nm, 1800–1975 nm and 2400–2500 nm). More details on the field and laboratory measurements are available in ten Den et al. (2022) and Prikaziuk et al. (2022).

2.3. Model description

2.3.1. Crop growth model WOFOST

The WOFOST model (Boogaard et al., 2014) is a mechanistic model that simulates daily crop growth based on the specific soil, crop, and meteorological conditions. In the current study, we employed the WOFOST version 8.0 potential engine, embedded within the Python package PCSE (De Wit, 2024). PCSE, developed by Allard de Wit, contains various crop model implementations, including the WOFOST model.

2.3.2. Radiative transfer model SCOPE RTMo

The Soil Canopy Observation, Photosynthesis and Energy fluxes (SCOPE) model (Yang et al., 2021a) is a radiative transfer model proficient in simulating reflectance in the optical domain (400–2500 nm) for soil, leaf, and canopy, as well as emitted radiance in the thermal domain for canopy and soil. Relevant modules of the SCOPE model, integrated to simulate top-of-canopy reflectance, as per the objectives of this study, include:

(i) The Brightness-Shape-Moisture (BSM) model, which computes soil reflectance utilizing four specific parameters (Yang et al., 2021b).
(ii) The Fluspect model, a leaf optical radiative transfer model, which calculates leaf reflectance, transmittance, and fluorescence based on leaf composition (Vilfan et al., 2016),
(iii) RTMo, the radiative transfer of incident solar radiation in the vegetation canopy module. RTMo is a numerical adaptation of the SAIL model (Verhoef, 1984) computing the optical properties of vegetation layers based on parameters such as LAI, leaf angle distribution (characterized by LIDFa and LIDFb parameters), and illumination-observation geometry (solar zenith angle, viewing zenith angle, and the azimuthal difference between illumination and view angles).

2.4. Coupling WOFOST and SCOPE for remote sensing data assimilation

Within the remote sensing data assimilation framework, WOFOST is linearly linked with SCOPE RTMo. The coupling of the two models is a multi-step process, involving the integration of various parameters and model outputs (Fig. 3). The resulting coupled model facilitates daily simulations of top-of-canopy reflectance at multiple time points. To refine the calibration of selected crop parameters, a custom cost

function combined with a nonlinear optimization algorithm is employed to minimize the difference between the simulated and measured reflectance time series.

The operation of the coupled model proceeds as follows: Initially, the WOFOST model is set up and executed. The results from this simulation serve as input parameters for the SCOPE model. As a result of this process, a time series of growth variables and their corresponding top of canopy reflectance are simulated. By re-initializing key parameters and re-running the coupled model, it becomes possible to simulate a varied set of growth variables and reflectance time series. As a next step, a measured reflectance time series — a sequence of reflectance data, collected over the growing period — is introduced. This data serves as a benchmark for calibrating the model's initial parameters. The model's initial parameters are systematically changed and the resulting simulated reflectance is compared against the measured. The objective of this process is to adjust the initial parameters until the model's output aligns closely with the actual measurements. To achieve the aforementioned process, which minimizes the difference between the simulated and measured reflectance, we employ data assimilation techniques, incorporating the use of a cost function and an optimization algorithm (described further in Section 2.4.3).

Because the version of the WOFOST model used in this study is coded in Python, whereas the radiative transfer model SCOPE is coded in Matlab, the MATLAB 2019b Engine for Python was used for the coupling and the data transfer between the two models. The assimilation process for remote sensing data is executed within the Jupyter Notebook Python interface. The next sections describe the coupling steps in more detail.

2.4.1. WOFOST simulation and state variable conversion

The integration process begins with initializing the WOFOST model, inputting weather conditions, measured in the field, genotypical crop parameters and agromanagement practices. WOFOST then proceeds to simulate the crop's growth throughout the cultivation period, outputting daily values of key variables such as Leaf Area Index (LAI), Dry Weight of Leaves (WLV), and Specific Leaf Area (SLA). While LAI, obtained from WOFOST, seamlessly integrates with SCOPE, WLV and SLA need to be converted into leaf dry matter (Cdm), and leaf mesophyll structure (N), respectively, before they can be used as inputs.

WLV, expressed as kilograms per hectare of ground area, requires transformation to grams of dry leaves per square centimeter of leaf area. This is done by utilizing the equation:

$$Cdm = \frac{WLV}{LAI} \cdot 10^5 \tag{1}$$

The estimation of leaf mesophyll structure (N) is achieved through the empirical relationship with SLA, as developed by Jacquemoud and Baret (1990):

$$N = \frac{0.025 + 0.9 \cdot SLA}{SLA - 0.1} \tag{2}$$

2.4.2. SCOPE simulation

To perform a SCOPE model simulation, a series of parameters should be assigned, for the model's modules (see 2.3.2). The soil parameters were determined through soil reflectance measurements taken alongside the canopy reflectance data. Solar zenith angle (tts) is calculated based on the latitude, longitude, and measurement time of the experimental fields, while the observation zenith angle (tto) is fixed at 0 degrees.

The chlorophyll content (Cab) was retrieved in a previous study (Prikaziuk et al., 2022), via SCOPE RTMo inversion, separately for each measurement day from the reflectance measurements and was provided to the coupled model independently. Chlorophyll content values ranged from 10 to 60 μg cm^{-2}, with an R^2 of 0.6 and RMSE of 6.16 μg cm^{-2}, compared to the measurements. Carotenoid content (Cca) was assumed to be 25% of the Cab value (SCOPE model default method (Van der Tol, 2024)).

Additionally, the Leaf Angle Distribution Function (LIDF) parameters, LIDFa and LIDFb, describe the cumulative leaf orientation distribution mathematically and are essential for characterizing the canopy structure. LIDFa determines the mean leaf orientation and LIDFb the bi-modality. Their values and also their sum and difference must lie in the interval [−1 1]. Because these parameters have a significant impact on the top-of-canopy reflectance simulation and a physical meaning, they are constrained to ensure specific canopy shapes, such as spherical, uniform, erectophile, or planophile.

2.4.3. Optimization

After following the steps outlined in the preceding two subsections, namely, conducting a simulation using the WOFOST model, transferring its output to the SCOPE model, and subsequently carrying out a SCOPE simulation, a time series of growth variables and their corresponding top of canopy reflectance, are simulated (hereafter Coupled model). The optimization's purpose is to detect the values of selected parameters, that minimize the difference between the measured and simulated top-of-canopy reflectance time series.

The coupled model optimization follows the logic of the re-calibration data assimilation technique (Jin et al., 2018) (Fig. 1a). Selected initial parameters are adjusted using a non-linear optimization algorithm. The first objective was to select parameters, related to individual modules of crop modeling, namely phenology, photosynthesis, partitioning and leaf expansion while, at the same time avoid selecting a big number that will make the objective of optimization impossible. According to the documentation, the source code of the corresponding modules and studies on optimization or callibration (ten Den et al., 2022; Dlamini et al., 2023), several parameters were identified that significantly influence crop growth and have a physical and physiological meaning:

(i) TDWI: Initial total crop dry weight.
(ii) TSUM1: Temperature sum from emergence to anthesis, or tuber initiation for potatoes.
(iii) TSUM2: Temperature sum from anthesis (or tuber initiation for potatoes) to maturity.
(iv) SPAN: life span of leaves growing at 35 °C
(v) AMAXTB: Maximum leaf assimilation as tabular parameter, based on the development stage.
(vi) SLATB: Specific leaf area based on the development stage. This is a tabular parameter with multiple values for different Developmental Stages (DVS). For potatoes, default SLATB values are set at 0.003 ha kg^{-1} from DVS 0.0 to 1.1, which then linearly decreases to 0.0015 at DVS 2.0. Recent research (ten Den et al., 2022)updated this to SLATB = 0.0017 ha kg^{-1} at DVS = 0.0 and 0.0024 ha kg^{-1} at DVS = 1.0 and 2.0. The updated values describe new varieties more accurately, compared to the old values.
(vii) FLTB, FSTB, FOTB: These parameters represent the fraction of total dry matter allocated to different organs. They are tabulated and contingent on DVS. Linear interpolation is employed between values at distinct DVS, and their cumulative values must equal 1 for every DVS. Specifically: FLTB (allocation to leaves) starts at 0.8 (80% of the produced dry matter) and it diminishes to 0. FSTB (allocation to stems) begins at 0.2 (20%) and reduces to 0. FOTB (allocation to storage organs) starts at 0, and escalates to 1 (100%). Recent research (ten Den et al., 2022) updated these parameters to variety specific values, to capture their crop growth more accurately.

Reducing the number of parameters is crucial for the optimization process, to avoid the problem of ill-posedness. Therefore, the SLATB was reduced to one value for optimization with the boundaries being the same as in ten Den et al. (2022), excluding the beginning and end of period, due to lack of reflectance measurements at these points. The three partitioning tabular parameters have been transformed into

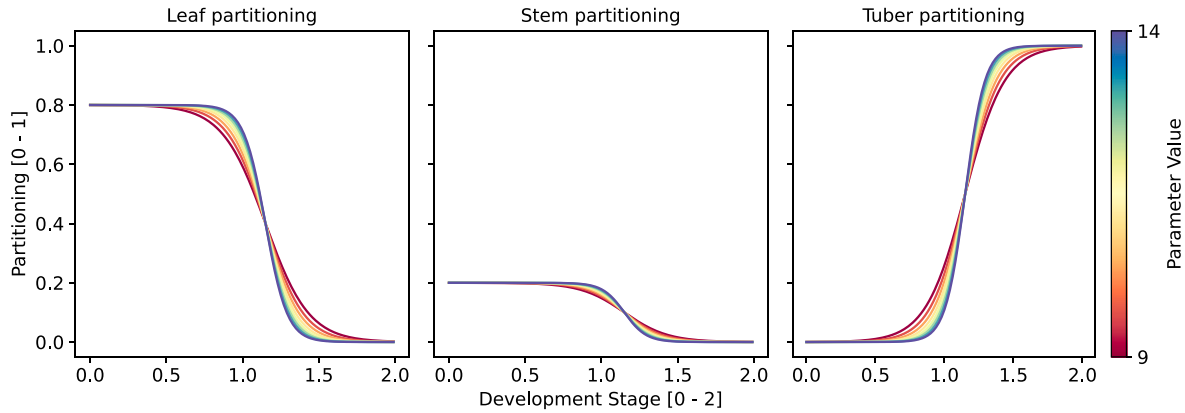

Fig. 4. Leaf (FLTB), stem (FSTB), tuber (FOTB) partitioning S-Curves. The figure shows the change of the slope of the curves.

three S-curves, parameterized by a single parameter, the S-curve slope (Fig. 4). The S-curves had a fixed centroid in order to keep the tuber partitioning curves close to the ones shown in ten Den et al. (2022), without adding a lot of parameters to the optimization process.

Based on the logic of the coupled model -WOFOST simulation, variable conversion, SCOPE simulation-, the aforementioned parameters are connected to the WOFOST simulation. After the conversion to SCOPE RTMo inputs, most of the SCOPE RTMo parameters and variables are being calculated or retrieved independantly, as mentioned in Section 2.4.2. For the SCOPE model the Cab variable was assigned based on an independent retrieval because of two reasons. While the canopy reflectance is sensitive to Chlorophyll, its effect is located to the region between 500 and 700 nm, where the combined effect of the rest of variables is similar or higher. Moreover, the Chlorophyll content can be retrieved with high accuracy and therefore it will not add too much error to the coupled model process. As a result, the water content (Cw) has been selected to be included in the optimization process because it greatly influences a large part of the reflectance from 1000 to 2400 nm and therefore false retrieval would result in greater error to both the LAI and as a consequence to the simulated reflectance. In total, six parameters were selected for WOFOST while one parameter, Cw, as an array of size equal to the number of measurement, for SCOPE RTMo. All the coupled model parameters, both selected for optimization and optimized in another manner are shown in Table 2

The optimization is facilitated by a cost function that serves as minimization target: It quantifies the difference between the time series of reflectance simulated by the coupled model and the measured one, plus a background (prior) contribution. The cost function was adopted from Van der Tol et al. (2016):

$$f = \left(\frac{B - B_0}{\sigma_B}\right)^T \cdot \left(\frac{B - B_0}{\sigma_B}\right) \cdot 0.03 + \sum_{i=1}^{t}\left(\left(R_i - R_{i_0}\right)^T \cdot \left(R_i - R_{i_0}\right)\right) \quad (3)$$

The cost function consists of two parts, the cost of the background (first term) and the cost of the observation (second term). In the equation above, B is the vector of the reinitialized parameters (Table 2), B_0 is the vector with prior information and σ_B is the standard deviation of these parameters. For this study, the background parameter (B_0) information was the midrange of the parameter range of the Table 2 while the first guess (initialization) of the vector B was set as the parameter set that is included in the PCSE/WOFOST download, namely "potato701" (PCSE, 2024). R_i is the simulated reflectance, the output of the coupled model with the parameters B. Finally, R_{i_0} is the measured reflectance. The letter T stands for transpose and i represents the index of the set of the observation days. The addition of background information makes the retrieval more stable by preventing unrealistic posterior values for the parameters, where the coefficient 0.03 ensures the background has a much smaller weight compared to the observations (Van der Tol et al., 2016). For the minimization of the cost function, the Python library NLopt and the SUBPLEX algorithm has been selected (Johnson, 2024). To examine the impact of initial WOFOST parameter values on the minimization process, a second round of optimization have been done with the first guess being the initial values calculated for the same varieties as described in the study of ten Den et al. (2022). The non-linear optimization algorithm (SUBPLEX) re-initializes the selected parameters, and the data assimilation procedure ends when the cost function is minimized. To ensure that minimization will be achieved and the optimization will not stop due to limited evaluation number, the number of evaluations, allowed by the optimization was set to 1500, while the minimum criteria for stopping the evaluation was set at 0.01.

The selection of the algorithm was based on its computational speed. Moreover, it is a derivative-free, local search algorithm that does not require the calculation of analytical gradients. This is important because calculating such gradients for a crop model is a very time-intensive task. After the minimization of the cost function, the values of the reinitialized parameters are used as the new inputs for the coupled WOFOST-SCOPE model to simulate the potato crop growth in question.

2.5. Global sensitivity analysis

The influence of the initial model parameters, on both the reflectance and the crop growth simulations, varies across different time periods and developmental stages. As a result, different parameters, or sets of parameters, will affect the simulations at different stages and with varied intensity. To visualize this impact, in both the temporal and spectral domains of the crop growth simulations, a global sensitivity analysis of the coupled WOFOST-SCOPE model was performed. The global sensitivity analysis employed the Fourier Amplitude Sensitivity Test (FAST) methodology, facilitated by the SALib library (Herman and Usher, 2017) in Python, version 3.7. The analysis encompassed crop growth simulations derived from 3150 distinct initial parameter sets. The sensitivity indices were normalized, ensuring their cumulative representation equated to 1 for each timestep. These identical parameter sets were subsequently integrated with the coupled model to visualize their impact on the simulated reflectance.

2.6. Model evaluation

Simulated and measured spectra of reflectance were compared to each other by calculating their average difference per band and the RMSE per variety for every measurement day,separately for each field.

The simulated values of LAI, WLV and yield were compared to the values obtained through destructive field measurements, by calculating the root mean-squared error (RMSE) and the coefficient of determination (R^2) as measure for the goodness-of-fit.

To evaluate the added value of the coupled WOFOST-SCOPE model, we compared the yield, LAI and WLV of our simulations to the values as described in ten Den et al. (2022).

Table 2
Input parameters of the coupled WOFOST-SCOPE model.

Parameter	Description	Unit	Source/range
WOFOST			
TDWI*	Total initial crop dry weight	kg ha^{-1}	[70 120]
TTSUM	Total temperature sum emergence - maturity	°C days	measurement
TSUM1*	Temperature sum emergence - tuber initiation	°C days	[150 300]
TSUM2	Temperature sum tuber initiation - maturity	°C days	TTSUM - TSUM1
SPAN*	Leaf life span, growing at 35 Celsius	days	[25 40]
SLATB*	Specific leaf area (as function of DVS)	ha kg^{-1}	[0.002 0.0035]
DVS_SLOPE*	Slope of the partitioning S-curves	–	[9 14]
AMAXTB*	Maximum assimilation rate (as function of DVS)	kg ha^{-1} hr^{-1}	[25 40]
SCOPE			
Cab	Chlorophyll content	μg cm^{-2}	RTMo Retrieval
Cca	Carotenoid content	μg cm^{-2}	25% of Cab
N	Leaf mesophyll structure	–	Based on SLATB
Cdm	Leaf dry matter	g cm^{-2}	Based on WLV
Cw*	Leaf water content per measurement	cm	[0.015 0.04]
BSMB	Model parameter for soil brightness	–	BSM Retrieval
BSMlat	Model soil parameter 'lat'	–	BSM Retrieval
BSMlon	Model soil parameter 'lon'	–	BSM Retrieval
SMC	Volumetric soil moisture content	–	BSM Retrieval
tts	Solar zenith angle	°	Measurement
tto	Observer zenith angle	°	0
lat	Location latitude	°N	Field location
lon	Location longitude	°E	Field location
LIDFa	Leaf Inclination	–	−1 ∨ 0 ∨ 1

The parameters with * start with their respective default value and are being retrieved based on the optimization routine and minimization of the Eq. (3).

3. Results

3.1. Global sensitivity analysis

Figs. 5 and 6 present the global sensitivity analysis of the coupled WOFOST-SCOPE model in the temporal and spectral domains, respectively. Within the temporal domain (Fig. 5), the influence of each input parameter on the simulated variables (WLV, LAI, Cdm) varies across different periods of the growing season. At the start, TDWI and SLATB (specific leaf area) are the most influential. TDWI represents the initial dry weight, accounting for about 50% of the LAI and almost 100% of the Weight of Leaves (WLV). However, its impact decreased as the season progresses. SLATB, which helps convert WLV to LAI, initially affects LAI by about 50%, growing to around 90% influence towards the middle of the season, and then decreases when the leaves stop growing. Since LAI, WLV, and leaf dry matter (Cdm) are interconnected, SLATB also greatly impacts Cdm (almost 100%) and WLV, maintaining an average influence of around 15% throughout the season.

SPAN, which represents the lifespan of the leaves, becomes more important in the second half of the season, influencing both LAI and WLV by up to 80% at the end of the season.

TSUM parameters (TSUM1 and TSUM2) indicate the length of different growth stages (time from emergence to tuber initiation, and from tuber initiation to maturity). The two parameters dictate when the crop moves to the next development stage and how the total biomass is partitioned to the different organs. TSUM1 affects the LAI by about 10% and WLV by about 30%. TSUM2 impacts the crop growth similarly.

The partitioning parameter (DVS_SLOPE) determines the biomass allocation to the different crop organs. Changing this partitioning parameter impacts the plant's growth by indicating when it starts forming tubers and how fast the biomass allocation changes. The impact of this parameter starts around the tuber initiation phase and increases over time up to the moment when the biomass mostly goes to the tubers.

AMAXTB changes the photosynthesis and therefore the biomass production of the crop and as a result, it affects the crop growth early in the season. The influence of this parameter is located mostly at the beginning with a pick around the time of the maximum LAI. From that moment onwards, its influence drops.

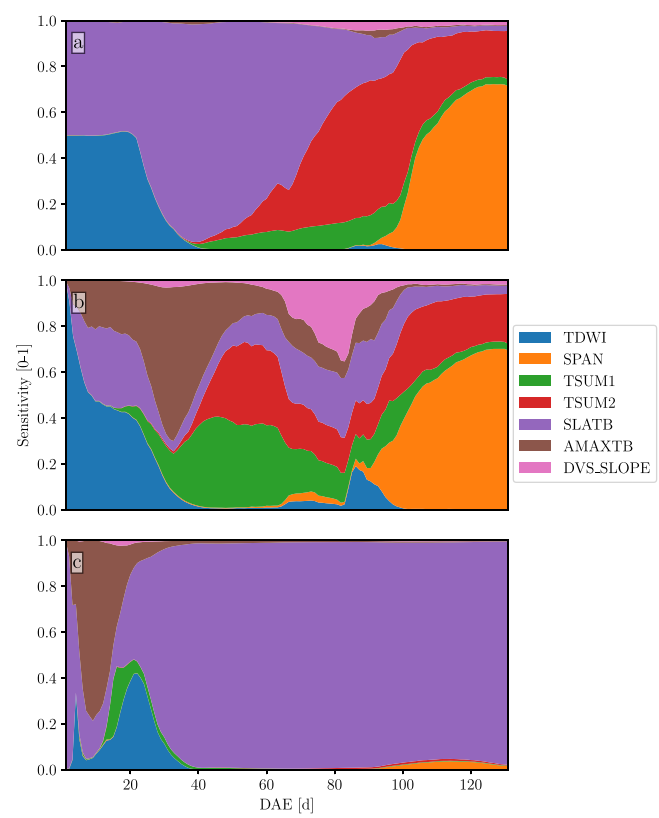

Fig. 5. Global sensitivity analysis of the selected parameters of the WOFOST model on the simulation of **a**: Leaf Area Index (LAI), **b**: Weight of Leaves (WLV) and **c**: Leaf dry matter (Cdm) over time. The *y*-axis refers to the percentage sensitivity (all the parameters are normalized to the sum of 1), and the *x*-axis refers to the days after emergence (DAE).

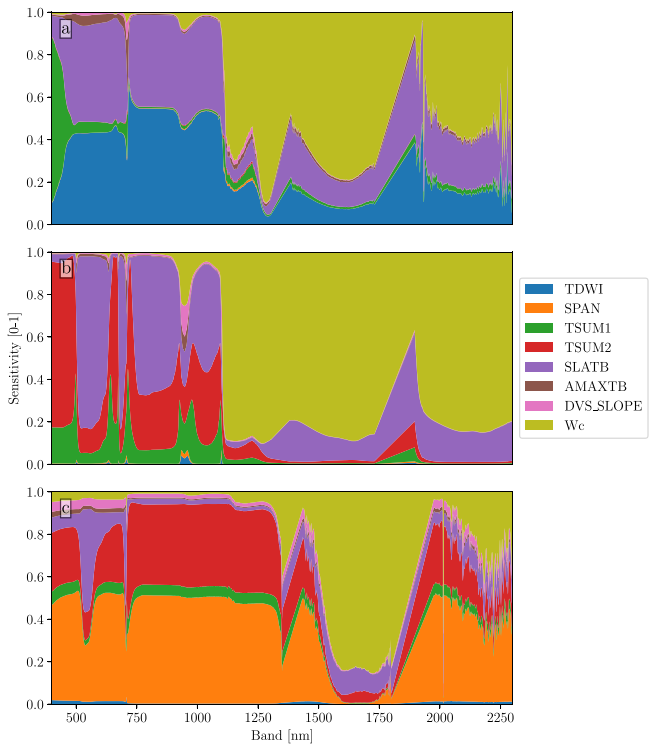

Fig. 6. Global sensitivity analysis of the coupled WOFOST - SCOPE model. Visualization of the impact of various WOFOST parameters on reflectance, captured at three crucial time points (days after emergence: **a**: 20, **b**: 60, **c**: 110). The *x*-axis displays the spectral bands in nanometers (nm), and the *y*-axis represents the total sensitivity index values of each parameter, normalized to sum up to 1.

The initial parameter set in WOFOST plays a critical role in shaping the simulated variables that the SCOPE model utilizes to simulate spectral reflectance. Hence, they indirectly influence the reflectance results. (Fig. 6).

At the beginning of the growing season, SLATB exerts a considerable impact on the simulated reflectance, accounting for more than 50% of the influence due to its dual impact on both LAI and Cdm. TDWI has a big influence while TSUM1 and AMAXTB also have a minor impact during this phase.

As the season advances, SLATB and TSUM1 continue to affect the results, and TSUM2 comes into play, while the partitioning parameter (DVS_SLOPE) has a small effect. Mid-season, the influence of the aforementioned parameters is most pronounced in the visible region of the spectrum, significantly less so in the NIR (Near-Infrared) and SWIR (Short-Wave Infrared) regions. This is because their impact is predominantly on dry matter, with a lesser effect on LAI due to saturation effects. Mid-season, the water content (Wc) dominates the effect in the SWIR region due to the radiation absorption by leaf water.

As the season progresses further, SLATB, TSUM2, and SPAN emerge as the parameters with the greatest impact on reflectance.

3.2. Assimilation of reflectance datasets into the WOFOST-SCOPE coupled model

We have assimilated measured time series of spectral reflectance datasets into the coupled WOFOST-SCOPE model and analyzed the discrepancies between the measured and simulated reflectance values. Fig. 7 illustrates the mean, standard deviation, and Root Mean Square Error (RMSE) of these differences.

The analysis of reflectance discrepancies reveals a pronounced variance within the spectral range of 700 to 1200 nm across both investigated locations. This phenomenon can be attributed to the varying sensitivity of reflectance to LAI and Cdm across the growing season. Both parameters are closely related to the SLATB parameter of WOFOST, which is shown in Fig. 6 to be one of the most influencing parameters. The SCOPE model, which employs LAI and Cdm among other plant traits to simulate reflectance, demonstrates that LAI, in particular, plays a crucial role in this context. The measured-simulated temporal reflectance discrepancy is higher later in the season because the influential variables (LAI, Cdm) exhibit temporal connectivity, with past values influencing later ones. RTMo prioritizes minimizing the reflectance difference by setting the underlying variable values. In this process, spectral measurements are treated individually without considering temporal realism. In contrast, the coupled model introduces temporal variable inter-connectivity constraint, which leads to higher simulated-measured reflectance difference compared to independent RTMo retrieval.

Furthermore, in this study, LAI exhibited overestimation in the majority of cases. However, the response of reflectance to LAI is not linear, particularly when LAI values approach or exceed the saturation point, typically around 3 to 4 m^2 m^{-2}. At this point, additional foliar layers contribute minimally to the outgoing radiation flux, resulting in a plateau in reflectance values. Consequently, the disparity between simulated and measured reflectance tends to be more pronounced for errors associated with simulated and actual LAI values, located below the saturation point compared to those closer to or above it. This is more apparent with the variety Premiere, where the last measurement was close to the day of maturity and therefore the error resulted from the differences between simulations and measurement at this point, exceeds significantly that of the rest of the cases.

3.3. Crop growth simulation, based on spectral reflectance and the coupled model

The comparison of the Leaf Area Index (LAI), Weight of Leaves (WLV), and yield between the measurements and the simulations performed by the WOFOST-SCOPE model, based on the reflectance assimilation, is a crucial aspect of our study, ensuring the reliability and accuracy of our modeling approach. Figs. 8–10 present a comparative analysis between the simulated values using spectral reflectance data assimilated into the coupled WOFOST-SCOPE model and the values obtained through destructive field measurements.

Fig. 8 specifically focuses on the evaluation of LAI. The simulation of LAI demonstrates varied levels of accuracy, contingent on the potato variety and field conditions. The coefficient of determination (R^2) for LAI simulations ranges from 57% to 68%, with the RMSE spanning from 1.09 to 1.34 m^2 m^{-2}. Specifically, the Markies variety exhibits the lowest correlation, with $R^2 = 0.57$ and RMSE = 1.34 m^2 m^{-2}. This is attributed to an overestimation of LAI by the model at the season's onset. Markies continued to grow throughout the season, reaching its peak LAI closer to maturity, around 100 days after emergence—a trend that the WOFOST model simulation failed to accurately capture, as it predicted the highest LAI value much earlier in the season.

Fontane L exhibited similar trends to Markies in simulating the highest LAI earlier than the measurements ($R^2 = 0.65$, RMSE = 1.22 m^2 m^{-2}). The Innovator variety showed a slight overestimation of seasonal LAI in the model simulation ($R^2 = 0.65$, RMSE = 1.25 m^2 m^{-2}). Fontane V displayed an underestimation of the maximum LAI ($R^2 = 0.63$, RMSE = 1.09 m^2 m^{-2}).

Festien showed a discrepancy in capturing seasonal LAI variation, resulting in overestimation of maximum LAI and minimal variation between simulations ($R^2 = 0.68$, RMSE = 1.17 m^2 m^{-2}). For Premiere, the simulation captured the seasonal LAI trend with an overestimation throughout the season ($R^2 = 0.62$, RMSE = 1.18 m^2 m^{-2}).

Fig. 9 provides a visual representation of the comparison between the simulated and the destructively measured Weight of Leaves (WLV).

In terms of WLV simulations, the trend observed was in alignment with that of LAI simulations, albeit with a notably lesser degree of

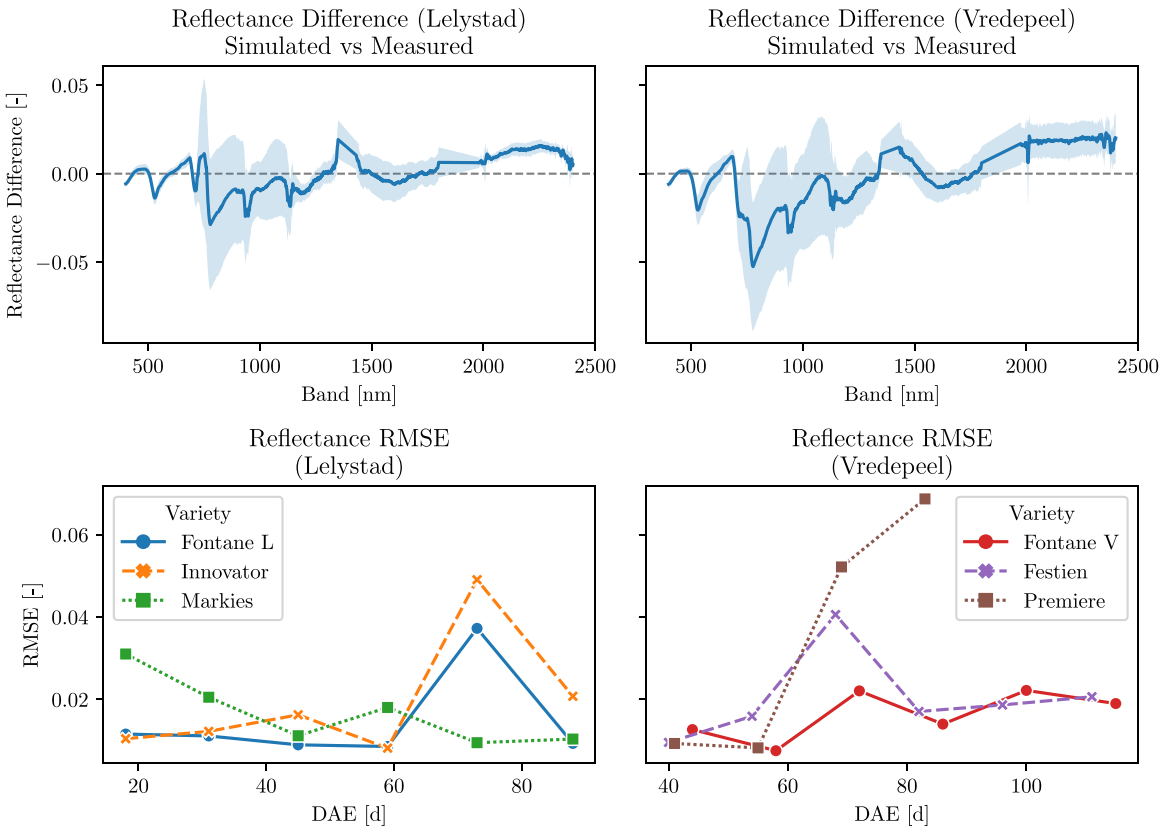

Fig. 7. Difference between simulated and measured reflectance. Top row shows the difference between the average measured and simulated spectral reflectance (mean and standard deviation of all the measurements). The bottom row shows the RMSE between measured and simulated spectral reflectance, per measurement day. The differences for the top row were calculated based on the total sample (three plots and six time-points per variety, except Premiere that had three plots and four time-points). For the bottom row, the average RMSE of three plots per variety and time point was used.

overestimation. The coefficient of determination for the WLV simulations ranged from 61% to 77%, with the RMSE varying between 310 and 440 kg ha^{-1}. Markies and Festien exhibited the lowest correlation (R^2 = 0.61 and 0.63, respectively) while Premiere had the highest error, followed by Markies (RMSE = 440 and 416 kg ha^{-1}, respectively). In Markies, the measured WLV increased over time, peaking around 100 days after emergence, a trend which was closer to the maturity stage. While the coupled model captured the initial increase in WLV, it struggled to follow the upward trend of the measurements.

Festien's simulation failed to capture their variations, resulting in the second lowest correlation. For Fontane V, the simulation aligned with measurements, but could not accurately replicate the WLV variation (R^2 = 0.7, RMSE = 383 m^2 m^{-2}). Fontane L showed similar behavior as Markies but with less error and overestimation (R^2 = 0.67, RMSE = 347 m^2 m^{-2}). The simulation of Innovator accurately captured the measurements (R^2 = 0.77, RMSE = 310 kg ha^{-1}). Premiere simulation closely followed the measurements, albeit with underestimation mid-season and overestimation towards the end (R^2 = 0.64, RMSE = 440 kg ha^{-1}).

Fig. 10 presents a comparison between the yield simulated using the coupled WOFOST-SCOPE model, following calibration based on spectral reflectance data, and the yield obtained through field measurements at the end of the growing season.

The results from the yield simulations exhibited a correlation of R^2 = 0.4 and RMSE = 1800 kg ha^{-1}, with the measured data. For the varieties Fontane V, Fontane L and Premiere, the simulations showed underestimation of the yield while for the varieties Festien, Markies and Innovator the simulations resulted in overestimation.

For further evaluation of our model, we compared the results for yield, leaf area index (LAI) and weight of leaves (WLV) obtained from the coupled SCOPE-WOFOST model to the values detailed in the work by ten Den et al. (2022) (hereinafter referred to as "TtD. '22"). The Fig. 11 (and Table B.5) visualizes the yield, LAI and WLV assessments of this study, "TtD. '22" and SCOPE retrieval (SCOPE cannot directly estimate yield).

The comparison for the yield predictions, revealed that our combined WOFOST-SCOPE model exhibited an average total deviation of 3% from the actual measurements. In contrast, the results from "TtD. '22" demonstrated a slightly higher average deviation of 4%.

When examining Leaf Area Index (LAI), the coupled WOFOST-SCOPE model outperformed "TtD. '22" for four out of six varieties, achieving an average relative Root Mean Square Error (rRMSE) that was 9% lower. Interestingly, the performance gap between WOFOST-SCOPE and the standalone SCOPE model was consistent, with an average difference of 6%, underscoring the comparability of their predictive accuracy.

In the evaluation of the WLV variable, the WOFOST-SCOPE model demonstrated superior performance compared to "TtD. '22" across three varieties, worse performance for one, and for two other varieties, the rRMSE values between WOFOST-SCOPE and "TtD. '22" were equivalent, showcasing consistent performance. Notably, the SCOPE retrieval method encountered significant challenges, with its rRMSE notably exceeding those of both WOFOST-SCOPE and "TtD. '22" by over 50%, underscoring a substantial discrepancy in accuracy.

The two optimization rounds, based on different initial crop parameter values, showed minor differences, with the average rRMSE between them being equal to 0.1 (rRMSE [LAI, TWSO]: 0.11, rRMSE [WLV]: 0.08). This comparison broadly demonstrates that the use of reflectance data, as opposed to traditional manual measurements, offers comparably precise results. This indicates a promising direction for enhancing accuracy and efficiency in agricultural modeling by leveraging advanced data collection techniques.

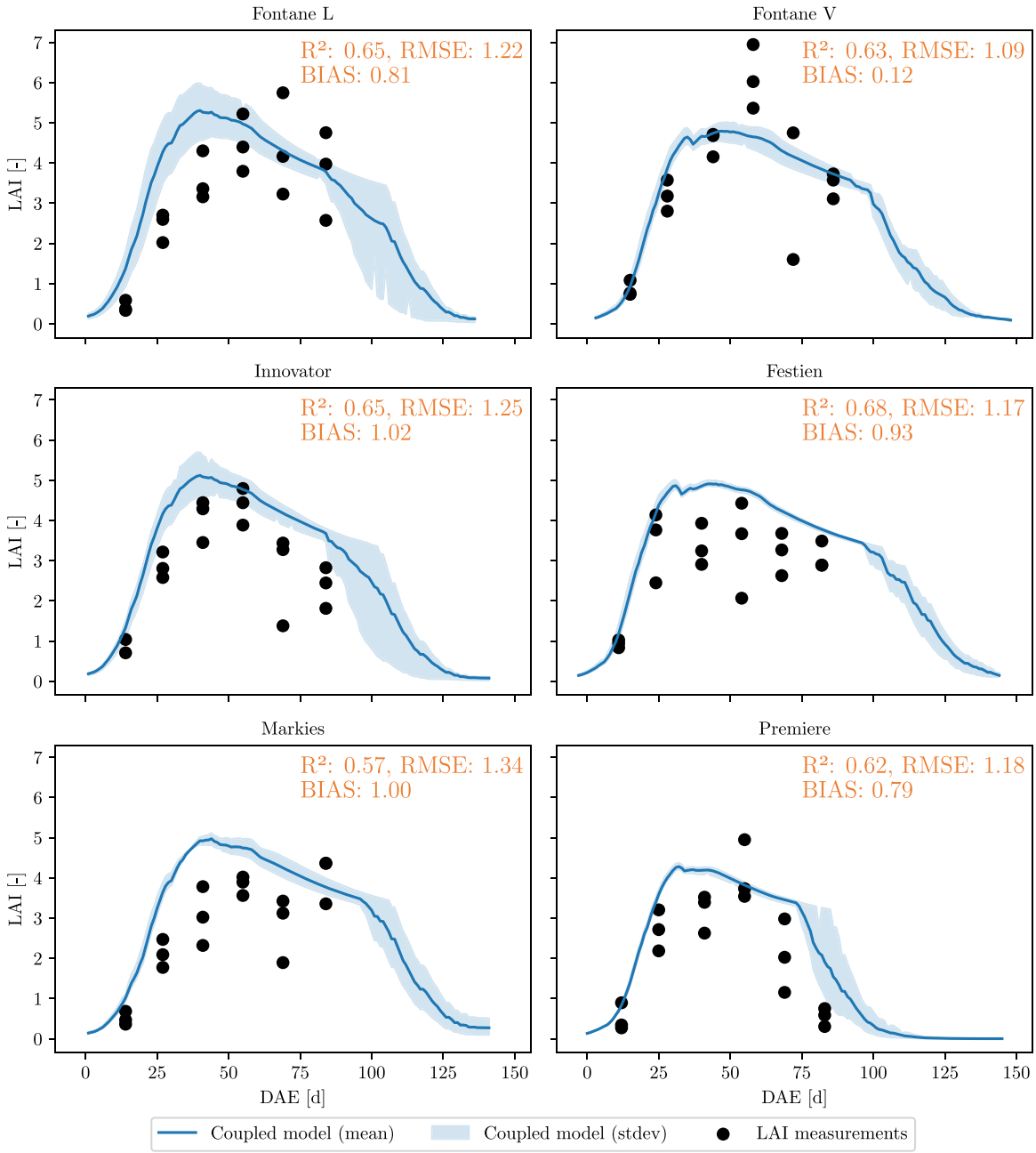

Fig. 8. Visualization of simulated LAI with coupled WOFOST-SCOPE model (Coupled model (mean)), the standard deviation based on simulations from the three plots (Coupled model (stdev)) and destructively measured LAI (LAI measurements). The orange text contains R2, RMSE and BIAS from linear regression between the Coupled model simulation and the measurements. V in Fontane V denotes location Vredepeel and L in Fontane L Lelystad.

4. Discussion

In this study, we have developed a coupled WOFOST-SCOPE model that uses spectral reflectance time series to simulate crop growth, marking a departure from traditional methods that depend on destructively measured variables like LAI for optimizing crop simulations (Huang et al., 2019a). Our method integrates output from the WOFOST model as hard constraints for the input parameters of the SCOPE radiative transfer model. By adjusting WOFOST model parameters and minimizing a cost function, we calibrate the model to closely match measured and simulated reflectance over time. We demonstrate that this approach produces comparable results for yield, leaf area index (LAI) and weight of leaves (WLV) as calibrated WOFOST simulations (ten Den et al., 2022), therefore enabling a potentially more accurate way to assimilate remote sensing data into crop growth modeling. The above is of course valid under conditions where reflectance measurements are available. When it comes to simulating crop growth to areas without available reflectance measurements, or extrapolate the optimized crop model to unseen fields and alternative time-periods, more research needs to be conducted.

Conventional data assimilation practices, where reflectance or Vegetation Indices (VIs) combined with retrieval algorithms are used, retrieve variables from measurements as isolated time points without acknowledging that the measurements belong to the same crop and should exhibit a coherent pattern reflective of the crop's temporal growth progress. A more dynamic approach, proposed in this study, combines crop models, driven by genotype and environmental parameters (GxE), to simulate crop growth. This generates crop growth variables reflecting the biological processes of the observed crop. These variables are then used by SCOPE to produce a time series of reflectance

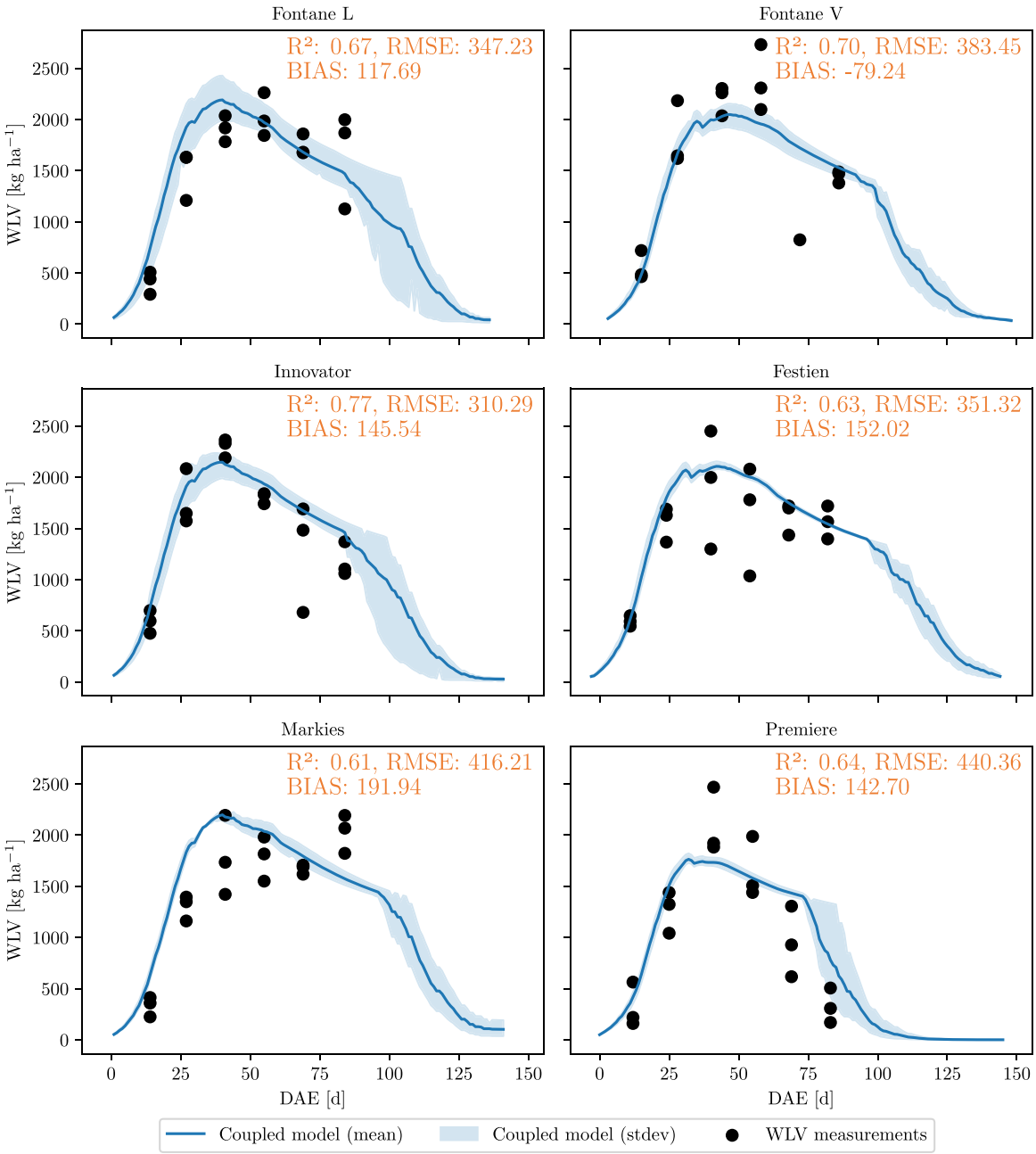

Fig. 9. Visualization of simulated WLV with coupled WOFOST-SCOPE model (Coupled model (mean)), the standard deviation based on simulations from the three plots (Coupled model (stdev)) and destructively measured WLV (wlv measurements). The orange text contains R2 and RMSE from linear regression between the Coupled model simulation and the measurements.

data tied to the crop's growth stages, establishing a coherent link between crop growth and reflectance. For example, in Prikaziuk et al. (2022), even though retrieval of LAI was successful ($R^2 > 0.5$), it is clear that, some values, mostly close to the maturity, have higher values than those at previous timesteps and thus, do not follow the expected temporal pattern of the potato growth cycle. In our study this issue was tackled by restricting the reflectance simulation, based on WOFOST growth simulations, although at the cost of higher RMSE in reflectance. Some of the outputs of the SCOPE model, such as carbon (GPP) and water (ET) fluxes, are not utilized in this study. ET assimilation is not possible because the WOFOST engine, in this study, did not have water stress dynamics. For GPP, SCOPE's parameters Vcmax and BBslope should be calibrated, before the simulation and the assimilation. This is done based on measurements such as eddy covariance which was not available. Alternatively it could be achieved based on the calibrated WOFOST which would result to an extra step without any clear benefit.

Using the coupled model introduces certain limitations. These include issues related to model simplifications, measurement or model errors, and the ill-posedness of inverse problems (Thorp et al., 2012). Furthermore, as seen from the value differences in Tables A.3 and A.4 the optimization may find value combinations and lead to better results, in some cases, but the trade-off is that the biological meaning of the parameters is reduced, compared to the destructive measurements, which conveys agronomic meaning to parameters. This trade-off allows the capturing of spatial variability and reduces the need for case specific parameter set.

Additionally, the model cannot replicate non-systematic errors inherent to plant trait measurements, leading to varying accuracy in daily

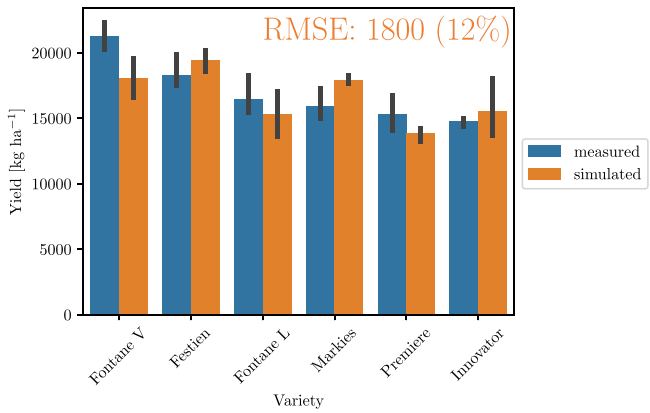

Fig. 10. Yield measurement per variety (meas) and coupled model yield simulation per variety (coupled).

reflectance simulations. On some days, the model's reflectance predictions closely match actual measurements, but on others, they diverge significantly. Despite this challenge, one of the notable strengths of our method lies in its ability to more effectively characterize and track uncertainties in reflectance, compared to retrieved state variables like LAI (Huang et al., 2019b). In addition, as described earlier in this section, by directly assimilating measured reflectance, we address the issue of retrieval continuity, acknowledging that reflectance measurements should not be treated as static, isolated points, but rather as a connected time series.

Furthermore, the optimization process focuses on a select group of parameters, meaning the model's accuracy might be compromised by the preset values of other parameters not included in the optimization, due to simplifications or to avoid ill-posedness. The success of our recalibration data assimilation strategy hinges on accurately identifying which WOFOST model parameters to adjust. WOFOST's initial parameters span various categories (crop, soil, weather and management) each influencing crop growth simulation differently and interacting with one another. It is important to note that within these, only crop-related parameters significantly impact the WOFOST model's predictions, as soil and management factors do not affect the potential growth simulation. Additionally, some parameters are vectors representing changes over the growth period, while others are singular values. Several studies such as Huang et al. (2019b), De Wit et al. (2012) and Curnel et al. (2011) selected key WOFOST crop parameters for re-initialization with various combinations of TDWI, TSUM1, TSUM2 and SPAN being the most popular, while AMAXTB, SLATB (specific leaf area) and partitioning parameters are identified as important based on a global sensitivity analysis (Ma et al., 2013a). In our study, we specifically selected TDWI, SPAN, TSUM, SLATB, AMAXTB, and a uniquely developed parameter—the slope of the partitioning curve—for re-initialization.

The global sensitivity analysis conducted in this study reveals that the selected parameters (each affecting crop growth simulations at various stages) were chosen for their significant impact across both temporal and spectral domains. Our analysis demonstrates how variations in Leaf Area Index (LAI) and Weight of Leaves (WLV) predominantly influence reflectance patterns over time, through specific leaf area or SLATB. This influence is expected, because both LAI and WLV are critical determinants of a plant's canopy structure and density, which in turn affect the canopy's ability to absorb and scatter light (Goudriaan and Monteith, 1990). The changes in these parameters alter the spectral signature of the vegetation simulated by SCOPE, as they modify the amount of light intercepted by the canopy and the efficiency of light utilization for photosynthesis. Such modifications are well captured by radiative transfer models, which simulate how light interacts with plant leaves and canopies. This relationship underscores the importance of accurately modeling LAI and WLV to predict reflectance accurately (Verrelst et al., 2015).

Our results reveal distinct peaks in the error between measured and simulated reflectance, consistently located in similar spectral regions across all cases. This pattern reinforces findings from previous research, notably a study reported by Berger et al. (2018), indicating a systematic occurrence of such discrepancies in reflectance simulation. Huang et al. (2019b), followed a similar methodology to assimilate satellite based reflectance and it shows similar results where the difference is bigger for the bands between 500 and 1300 nm, followed by the bands higher than 1300 nm, with the bands between 450–500 nm showing the lowest difference. For all regions, the difference in the aforementioned study is higher than ours. This could be attributed to the difference in scale (satellite vs in-field reflectance measurements) as well as that MODIS has low spatial resolution which means that the pixels could have information not related to crops. Furthermore, we observe differences in the error over the temporal domain. These errors are associated with simulated LAI values below the saturation point. We see that the last point in Premiere exceeds the error of the rest. This can be explained by the LAI difference being large and at the same time the LAI values being located below the saturation point of about 3–4 m^2 m^{-2}. Of course, LAI is not the only reason that can cause such errors with LIDF, Cdm and Cw playing role in it as well. Especially LIDF, where SCOPE retrieval optimizes the value so the total reflectance RMSE is minimized, the coupled model uses one value for the whole season which results in higher error at some points. Moreover, we observed a temporal lag between measured and simulated LAI, a phenomenon also documented by Novelli et al. (2019) in their work with the EPIC model for winter wheat using remote sensing data assimilation. This temporal lag underscores the challenges in accurately synchronizing simulated growth metrics with actual measurements.

Comparative analysis with existing literature further illustrates the variability in LAI simulation accuracy across different models and crops. For instance, the STICS crop model, as applied to corn, soybean, and spring wheat using a re-initialization data assimilation strategy, achieved LAI correlations ranging from 0.57 to 0.92 according to Jégo et al. (2012). Similarly, the LINTUL-3 model, calibrated with diverse sensor data for potato crops, exhibited LAI correlations from R2 = 0.31 to 0.61 and RMSE from 1.04 to 1.83 m^2 m^{-2} (Rijk, 2013).

Research on re-calibration of crop models using remote sensing data for potatoes is scarce, yet numerous studies have been conducted on yield estimation improvements for other crops. These studies, primarily focused on winter wheat Ma et al. (2013a) and Huang et al. (2015), have reported yield estimation correlations (R2) ranging from 0.28 to 0.77 and RMSE values between 373 to 889 kg ha^{-1}. Notably, Huang et al. (2019b) stands out as the only other research assimilating reflectance data directly into the WOFOST model, achieving a yield simulation correlation of R2 between 0.25 and 0.52 and RMSE between 710 kg ha^{-1} and 803 kg ha^{-1}.

In our study, we observed similar trends with a yield R2 of 0.4 and an rRMSE of 12%. The discrepancy in RMSE, particularly our higher values, can be attributed to the differences in the crops studied. While the aforementioned studies largely concentrated on crops like wheat, which generally have lower production levels, our focus on potatoes (a crop with typically higher yields) naturally results in larger RMSE values. This distinction underscores the impact of crop type on the expected accuracy and reliability of yield simulations derived from remote sensing data assimilation.

Our study has shown that the assimilation of reflectance data, rather than direct biophysical parameters like LAI, offers a viable route for optimizing crop models. A future study should be focused on the assimilation of satellite observed reflectance in the coupled model. Moving from within field to satellite scale, calls for attention on challenges such as long time periods without observations, due to cloudy conditions, pixels containing mixture of crops and of other land

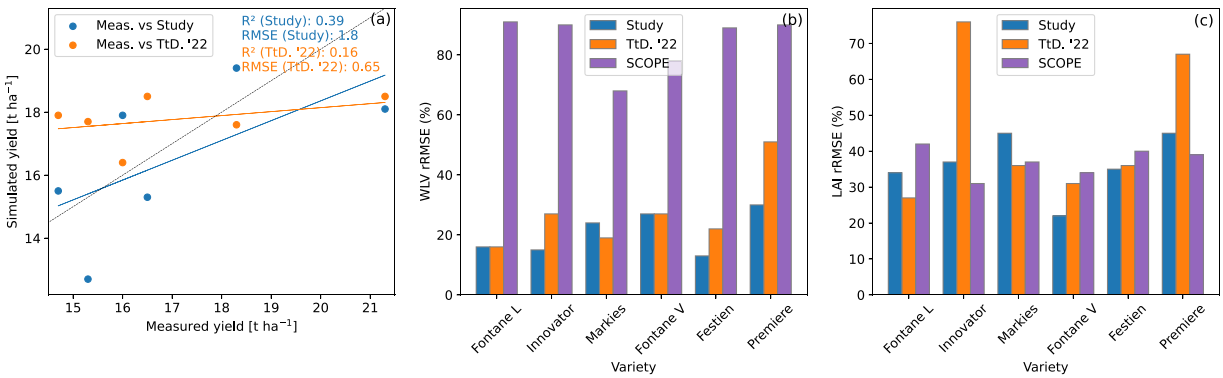

Fig. 11. Comparative analysis of simulation accuracy for yield, Leaf Area Index (LAI), and Weight of Leaves (WLV) across three distinct approaches: (a) scatter plot with two correlation lines: [Meas. vs Study] shows the correlation between measured yield and the coupled SCOPE-WOFOST. [Meas. vs. TtD. '22] shows the correlation between measured and the TtD. '22 study (ten Den et al., 2022). (b & c) Barplots show the rRMSE between this study, TtD. '22 and SCOPE retrieval for LAI and WLV respectively. The table with the data is in Table B.5.

uses (Huang et al., 2019b) and phenology shifts (Curnel et al., 2011). Moreover, we recommend expanding the application of this strategy across broader spatial extents, over multiple years, and among diverse crops and varieties to fully assess its potential. Previous studies, such as those by Huang et al. (2019b) and De Wit et al. (2012), have explored WOFOST model optimization through remote sensing data assimilation over several years, noting variable performance. Our findings similarly indicate that the success of data assimilation varies by potato crop variety and field, suggesting that factors like cultivar selection, environmental conditions, agricultural management practices, and the extent of temporal and spatial coverage may introduce uncertainties not fully accounted for by the current model and calibration approach.

5. Conclusions

This study represents a step forward in agricultural science, showcasing the effective combination of the WOFOST crop growth model and the SCOPE radiative transfer model. This combination was developed to enhance crop growth management and make yield predictions more accurate through the use of time-series, remotely sensed reflectance data. In our approach, the linkage between the two models is based on reflectance values, instead of state variables, and the results are comparable to, and in some instances better than, traditionally calibrated WOFOST simulations, particularly for yield, LAI and WLV predictions.

We suggest broader application of this method across different spatial and temporal scales, and for a variety of crops, to validate its wider applicability. While variability in performance across crop types and environmental conditions points to the complexity of agricultural modeling, our research indicates a promising path forward.

Future work should investigate alternative data assimilation techniques to reduce computational load and further explore the model's application under various abiotic stress conditions. The positive interplay observed between crop and radiative transfer models in our study hints at the potential for their full integration, possibly by using WOFOST methods such as partitioning and organ dynamics, directly with SCOPE, promising better crop monitoring and management strategies.

CRediT authorship contribution statement

Georgios Ntakos: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Egor Prikaziuk:** Writing – review & editing, Visualization, Resources, Investigation, Data curation. **Tamara ten Den:** Writing – review & editing, Resources, Methodology. **Pytrik Reidsma:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Nastassia Vilfan:** Writing – review & editing, Visualization, Methodology. **Tamme van der Wal:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Christiaan van der Tol:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors thank ITC GeoScience Lab personnel, Dr. Caroline Lievens, Kathrin Zweers-Peter and Camilla Marcatelli, WUR researcher Inge van de Wiel for the organization of the field work, Open Teelten Lelystad and Proefboerderij Vredepeel for lending the experimental fields.

Funding

EP, GN, TvdW and CvdT has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie grant agreement No 721995. Funding for experiments and TtD and PR came from the project "Yield gap analysis for sustainable potato production (Potato Gap NL)", project number 16891, funded by NWO-TTW and Holland Innovative Potato.

Appendix A. Initial parameter value difference

See Tables A.3 and A.4.

Appendix B. Summarized values for yield, LAI rRMSE and WLV rRMSE from three studies

See Table B.5.

Table A.3
Comparison of the Initial parameter values. "Study" refers to the average values, found in this study, based on the coupled model and the cost function minimization. "TtD. '22" are the parameters, calculated in ten Den et al. (2022), based on destructive measurements.

	TSUM2 (°Cd)	SLATB0 (ha kg^{-1})	SLATB1	SLATB2	AMAX (kg ha^{-1} hr^{-1})
Study	1650	0.0027	0.0027	0.0027	32
TtD. '22	2100	0.0017	0.0024	0.0024	33

Table A.4
Comparison between the variety specific initial parameter values in ten Den et al. (2022) and the average values of this study.

Variety specific	TSUM1 (°Cd)		SPAN (d)	
	Study	TtD. '22	Study	TtD. '22
Fontane L	238	222	31	32
Innovator	209	187	31	35
Markies	189	257	33	37
Fontane V	199	222	34	32
Festien	222	280	37	32
Premiere	180	164	29	35

Table B.5
Comparative analysis of simulation accuracy for yield, Leaf Area Index (LAI), and Weight of Leaves (WLV) across three distinct approaches: (1) the coupled SCOPE-WOFOST model developed in this study (hereafter referred to as "Study"), (2) the WOFOST simulations as detailed in the study by ten Den et al. (2022) (labeled as "TtD. '22"), and (3) the retrieval of LAI and WLV from reflectance data using the standalone SCOPE model (Prikaziuk et al., 2022) (referred to as "SCOPE"). Note that yield is not retrievable by the SCOPE model.

Variety	Yield [t ha^{-1}]			LAI rRMSE (%)			WLV rRMSE (%)		
	Meas.	Study	TtD. '22	Study	TtD. '22	SCOPE	Study	TtD. '22	SCOPE
Fontane L	16.5	15.3	18.5	34	27	42	16	16	91
Innovator	14.7	15.5	17.9	37	76	31	15	27	90
Markies	16	17.9	16.4	45	36	37	24	19	68
Fontane V	21.3	18.1	18.5	22	31	34	27	27	78
Festien	18.3	19.4	17.6	35	36	40	13	22	89
Premiere	15.3	12.7	17.7	45	67	39	30	51	90

References

Akhavizadegan, F., Ansarifar, J., Wang, L., Huber, I., Archontoulis, S.V., 2021. A time-dependent parameter estimation framework for crop modeling. Sci. Rep. 11 (1), 1–15. http://dx.doi.org/10.1038/s41598-021-90835-x, 2021 11:1.

Atzberger, C., 2013. Advances in remote sensing of agriculture: Context description, existing operational monitoring systems and major information needs. Remote Sens. 5 (2), 949–981. http://dx.doi.org/10.3390/rs5020949.

Berger, K., Atzberger, C., Danner, M., Wocher, M., Mauser, W., Hank, T., 2018. Model-based optimization of spectral sampling for the retrieval of crop variables with the PROSAIL model. Remote Sens. 10 (12), http://dx.doi.org/10.3390/rs10122063.

Boogaard, H.L., De Wit, A.J.W., Te Roller, J.A., Van Diepen, C.A., Rötter, R.P., Cabrera, J.M.C.A., Van Laar, H.H., Diepen, C.A.V., 2014. User's Guide for the WOFOST Control Centre 2.1 and WOFOST 7.1.7 Crop Growth Simulation Model. Technical Report.

Ceglar, A., van der Wijngaart, R., de Wit, A., Lecerf, R., Boogaard, H., Seguini, L., van den Berg, M., Toreti, A., Zampieri, M., Fumagalli, D., Baruth, B., 2019. Improving WOFOST model to simulate winter wheat phenology in Europe: Evaluation and effects on yield. Agricult. Sys. 168, 168–180. http://dx.doi.org/10.1016/J.AGSY.2018.05.002.

Cooper, M., Messina, C.D., 2022. Breeding crops for drought-affected environments and improved climate resilience. Plant Cell 35 (1), 162–186. http://dx.doi.org/10.1093/plcell/koac321.

Curnel, Y., de Wit, A.J., Duveiller, G., Defourny, P., 2011. Potential performances of remotely sensed LAI assimilation in WOFOST model based on an OSS Experiment. Agricult. Forest. Meterol. 151 (12), 1843–1855. http://dx.doi.org/10.1016/j.agrformet.2011.08.002.

De Wit, A., 2024. Python crop simulation environment. URL https://github.com/ajwdewit/pcse. (Accessed 20 January 2024).

De Wit, A., Duveiller, G., Defourny, P., 2012. Estimating regional winter wheat yield with WOFOST through the assimilation of green area index retrieved from MODIS observations. Agricult. Forest. Meterol. 164, 39–52. http://dx.doi.org/10.1016/j.agrformet.2012.04.011.

Dlamini, L., Crespo, O., van Dam, J., Kooistra, L., 2023. A global systematic review of improving crop model estimations by assimilating remote sensing data: Implications for small-scale agricultural systems. Remote Sens. 15 (16), http://dx.doi.org/10.3390/rs15164066, URL https://www.mdpi.com/2072-4292/15/16/4066.

Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., Hoersch, B., Isola, C., Laberinti, P., Martimort, P., Meygret, A., Spoto, F., Sy, O., Marchese, F., Bargellini, P., 2012. Sentinel-2: Esa's optical high-resolution mission for GMES operational services. Remote Sens. Environ. 120, 25–36. http://dx.doi.org/10.1016/j.rse.2011.11.026, The Sentinel Missions - New Opportunities for Science.

E.S.A., 2024. FLEX - earth online. URL https://earth.esa.int/eogateway/missions/flex, (Accessed 15 January 2024).

Faye, B., Webber, H., Gaiser, T., Müller, C., Zhang, Y., Stella, T., Latka, C., Reckling, M., Heckelei, T., Helming, K., Ewert, F., 2023. Climate change impacts on European arable crop yields: Sensitivity to assumptions about rotations and residue management. Eur. J. Agron. 142, 126670. http://dx.doi.org/10.1016/j.eja.2022.126670.

Goudriaan, J., Monteith, J.L., 1990. A mathematical function for crop growth based on light interception and leaf area expansion. Ann. Botany 66 (6), 695–701, arXiv:https://academic.oup.com/aob/article-pdf/66/6/695/348573/66-6-695.pdf. URL https://doi.org/10.1093/oxfordjournals.aob.a088084.

Guanter, L., Kaufmann, H., Segl, K., Foerster, S., Rogass, C., Chabrillat, S., Kuester, T., Hollstein, A., Rossner, G., Chlebek, C., Straif, C., Fischer, S., Schrader, S., Storch, T., Heiden, U., Mueller, A., Bachmann, M., Mühle, H., Müller, R., Habermeyer, M., Ohndorf, A., Hill, J., Buddenbaum, H., Hostert, P., Van der Linden, S., Leitão, P.J., Rabe, A., Doerffer, R., Krasemann, H., Xi, H., Mauser, W., Hank, T., Locherer, M., Rast, M., Staenz, K., Sang, B., 2015. The EnMAP spaceborne imaging spectroscopy mission for earth observation. Remote Sens. 7 (7), 8830–8857. http://dx.doi.org/10.3390/rs70708830.

Herman, J., Usher, W., 2017. SALib: An open-source Python library for Sensitivity Analysis. J. Open Sour. Softw. 2 (9), 97. http://dx.doi.org/10.21105/joss.00097.

Holzworth, D.P., Snow, V., Janssen, S., Athanasiadis, I.N., Donatelli, M., Hoogenboom, G., White, J.W., Thorburn, P., 2015. Agricultural production systems modelling and software: Current status and future prospects. Environ. Modell. Softw. 72, 276–286. http://dx.doi.org/10.1016/j.envsoft.2014.12.013.

Huang, J., Gómez-Dans, J.L., Huang, H., Ma, H., Wu, Q., Lewis, P.E., Liang, S., Chen, Z., Xue, J.H., Wu, Y., Zhao, F., Wang, J., Xie, X., 2019a. Assimilation of remote sensing into crop growth models: Current status and perspectives. In: Agricultural and Forest Meteorology. 276–277, http://dx.doi.org/10.1016/j.agrformet.2019.06.008.

Huang, J., Ma, H., Sedano, F., Lewis, P., Liang, S., Wu, Q., Su, W., Zhang, X., Zhu, D., 2019b. Evaluation of regional estimates of winter wheat yield by assimilating three remotely sensed reflectance datasets into the coupled WOFOST–PROSAIL model. Eur. J. Agron. 102, 1–13. http://dx.doi.org/10.1016/j.eja.2018.10.008.

Huang, J., Ma, H., Su, W., Zhang, X., Huang, Y., Fan, J., Wu, W., 2015. Jointly assimilating MODIS LAI and et products into the SWAP model for winter wheat yield estimation. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 8 (8), 4060–4071. http://dx.doi.org/10.1109/JSTARS.2015.2403135.

Jacquemoud, S., Baret, F., 1990. PROSPECT: A model of leaf optical properties spectra. Remote Sens. Environ. 34 (2), 75–91. http://dx.doi.org/10.1016/0034-4257(90)90100-Z.

Jacquemoud, S., Verhoef, W., Baret, F., Bacour, C., Zarco-Tejada, P.J., Asner, G.P., François, C., Ustin, S.L., 2009. PROSPECT + SAIL models: A review of use for

vegetation characterization. Remote Sens. Environ. 113 (SUPPL. 1), S56–S66. http://dx.doi.org/10.1016/j.rse.2008.01.026.

Jégo, G., Pattey, E., Liu, J., 2012. Using Leaf Area Index, retrieved from optical imagery, in the STICS crop model for predicting yield and biomass of field crops. Field Crops Res. 131, 63–74. http://dx.doi.org/10.1016/j.fcr.2012.02.012.

Jin, X., Kumar, L., Li, Z., Feng, H., Xu, X., Yang, G., Wang, J., 2018. A review of data assimilation of remote sensing and crop models. In: European Journal of Agronomy. Vol. 92, 141–152. http://dx.doi.org/10.1016/j.eja.2017.11.002.

Johnson, S.G., 2024. The nlopt nonlinear-optimization package. URL http://github.com/stevengj/nlopt, (Accessed 15 January 2024).

Kaplan, G., Fine, L., Lukyanov, V., Malachy, N., Tanny, J., Rozenstein, O., 2023. Using Sentinel-1 and Sentinel-2 imagery for estimating cotton crop coefficient, height, and Leaf Area Index. Agricult. Water. Manag. 276, 108056. http://dx.doi.org/10.1016/j.agwat.2022.108056.

Kasampalis, D., Alexandridis, T., Deva, C., Challinor, A., Moshou, D., Zalidis, G., 2018. Contribution of remote sensing on crop models: A review. J. Imaging 4 (4), 52. http://dx.doi.org/10.3390/jimaging4040052.

Launay, M., Guerif, M., 2005. Assimilating remote sensing data into a crop model to improve predictive performance for spatial applications. Agricult. Ecosys. Environ. 111 (1–4), 321–339. http://dx.doi.org/10.1016/j.agee.2005.06.005.

Li, M., Shamshiri, R.R., Weltzien, C., Schirrmann, M., 2022. Crop monitoring using Sentinel-2 and UAV multispectral imagery: A comparison case study in northeastern Germany. Remote Sens. 14 (17), http://dx.doi.org/10.3390/rs14174426.

Li-Cor, 2024. LI-2200C Plant Canopy Analyzer. URL https://www.licor.com/env/products/leaf_area/LAI-2200C, (Accessed 14 October 2023).

Ma, G., Huang, J., Wu, W., Fan, J., Zou, J., Wu, S., 2013a. Assimilation of MODIS-LAI into the WOFOST model for forecasting regional winter wheat yield. Math. Comput. Modelling 58 (3–4), 634–643. http://dx.doi.org/10.1016/j.mcm.2011.10.038.

Ma, H., Huang, J., Zhu, D., Liu, J., Su, W., Zhang, C., Fan, J., 2013b. Estimating regional winter wheat yield by assimilation of time series of HJ-1 CCD NDVI into WOFOST-ACRM model with Ensemble Kalman Filter. Math. Comput. Modelling 58 (3–4), 759–770. http://dx.doi.org/10.1016/j.mcm.2012.12.028.

Nicoleta Tantalaki, S.S., Roumeliotis, M., 2019. Data-driven decision making in precision agriculture: The rise of big data in agricultural systems. J. Agric. Food Inf. 20 (4), 344–380. http://dx.doi.org/10.1080/10496505.2019.1638264.

Novelli, F., Spiegel, H., Sandén, T., Vuolo, F., 2019. Assimilation of Sentinel-2 leaf area index data into a physically-based crop growth model for yield estimation. Agronomy 9 (5), 255. http://dx.doi.org/10.3390/agronomy9050255.

Panalytical, 2023. ASD FieldSpec 4 by malvern panalytical. URL https://www.malvernpanalytical.com/en/products/product-range/asd-range/fieldspec-range/fieldspec-4-standard-res-spectroradiometer, (Accessed 14 October 2023).

PCSE, 2024. PCSE: The python crop simulation environment — crop parameters for the WOFOST model. URL https://www.github.com/ajwdewit/WOFOST_crop_parameters/blob/master/potato.yaml, (Accessed 14 February 2024).

PhotosynQ, 2024. MultispeQ, PhotosynQ. URL https://www.photosynq.com/product-page/multispeq-v-2-0, (Accessed 15 January 2024).

Prikaziuk, E., Ntakos, G., ten Den, T., Reidsma, P., van der Wal, T., van der Tol, C., 2022. Using the SCOPE model for potato growth, productivity and yield monitoring under different levels of nitrogen fertilization. Int. J. Appl. Earth Obs. Geoinf. 114, 102997. http://dx.doi.org/10.1016/j.jag.2022.102997.

Qader, S.H., Utazi, C.E., Priyatikanto, R., Najmaddin, P., Hama-Ali, E.O., Khwarahm, N.R., Tatem, A.J., Dash, J., 2023. Exploring the use of Sentinel-2 datasets and environmental variables to model wheat crop yield in smallholder arid and semi-arid farming systems. Sci. Total Environ. 869, 161716. http://dx.doi.org/10.1016/j.scitotenv.2023.161716.

Radoglou-Grammatikis, P., Sarigiannidis, P., Lagkas, T., Moscholios, I., 2020. A compilation of UAV applications for precision agriculture. Comput. Netw. 172, 107148. http://dx.doi.org/10.1016/j.comnet.2020.107148.

Reidsma, P., Wolf, J., Kanellopoulos, A., Schaap, B., Mandryk, M., Verhagen, J., Ittersum, M.K., 2015. Climate change impact and adaptation research requires farming systems analysis and integrated assessment: A case study in the netherlands. Environ. Res. Lett. 10, 045004. http://dx.doi.org/10.1088/1748-9326/10/4/045004.

Rijk, B., 2013. Integration of Sensor Data in Crop Models for Precision Agriculture. Towards a Farmer's Decisopm Support System. Technical Report, Wageningen University and Research Centre, Laboratory of Geo-Information Science and Remote Sensing.

Seidel, S., Palosuo, T., Thorburn, P., Wallach, D., 2018. Towards improved calibration of crop models – Where are we now and where should we go? Eur. J. Agron. 94, 25–35. http://dx.doi.org/10.1016/j.eja.2018.01.006.

Shen, Y., Zhang, X., Yang, Z., Ye, Y., Wang, J., Gao, S., Liu, Y., Wang, W., Tran, K.H., Ju, J., 2023. Developing an operational algorithm for near-real-time monitoring of crop progress at field scales by fusing harmonized landsat and Sentinel-2 time series with geostationary satellite observations. Remote Sens. Environ. 296, 113729. http://dx.doi.org/10.1016/j.rse.2023.113729.

Siad, S.M., Iacobellis, V., Zdruli, P., Gioia, A., Stavi, I., Hoogenboom, G., 2019. A Review of Coupled Hydrologic and Crop Growth Models. Vol. 224, Elsevier B.V., 105746. http://dx.doi.org/10.1016/j.agwat.2019.105746.

Silva, J.V., Giller, K.E., 2020. Grand challenges for the 21st century: what crop models can and can't (yet) do. J. Agric. Sci. 158 (10), 794–805. http://dx.doi.org/10.1017/S0021859621000150.

ten Den, T., van de Wiel, I., de Wit, A., van Evert, F.K., van Ittersum, M.K., Reidsma, P., 2022. Modelling potential potato yields: Accounting for experimental differences in modern cultivars. Eur. J. Agron. 137, 126510. http://dx.doi.org/10.1016/j.eja.2022.126510.

Thorp, K.R., Wang, G., West, A.L., Moran, M.S., Bronson, K.F., White, J.W., Mon, J., 2012. Estimating crop biophysical properties from remote sensing data by inverting linked radiative transfer and ecophysiological models. Remote Sens. Environ. 124, 224–233. http://dx.doi.org/10.1016/j.rse.2012.05.013.

Van der Tol, C., Rossini, M., Cogliati, S., Verhoef, W., Colombo, R., Rascher, U., Mohammed, G., 2016. A model and measurement comparison of diurnal cycles of sun-induced chlorophyll fluorescence of crops. Remote Sens. Environ. 186, 663–677. http://dx.doi.org/10.1016/j.rse.2016.09.021.

Tripathy, R., Chaudhari, K.N., Mukherjee, J., Ray, S.S., Patel, N.K., Panigrahy, S., Singh Parihar, J., 2013. Forecasting wheat yield in Punjab state of India by combining crop simulation model WOFOST and remotely sensed inputs. Remote Sens. Lett. 4 (1), 19–28. http://dx.doi.org/10.1080/2150704X.2012.683117.

Van der Tol, C., 2024. SCOPE model github repository. URL https://github.com/Christiaanvandertol/SCOPE, (Accessed 15 January 2024).

Van Ittersum, M.K., Cassman, K.G., Grassini, P., Wolf, J., Tittonell, P., Hochman, Z., 2013. Yield gap analysis with local to global relevance—A review. Field Crops Res. 143, 4–17. http://dx.doi.org/10.1016/j.fcr.2012.09.009, URL https://www.sciencedirect.com/science/article/pii/S037842901200295X, Crop Yield Gap Analysis – Rationale, Methods and Applications.

Verhoef, W., 1984. Light scattering by leaf layers with application to canopy reflectance modeling: The SAIL model. Remote Sens. Environ. 16 (2), 125–141. http://dx.doi.org/10.1016/0034-4257(84)90057-9.

Verrelst, J., Rivera, J.P., van der Tol, C., Magnani, F., Mohammed, G., Moreno, J., 2015. Global sensitivity analysis of the SCOPE model: What drives simulated canopy-leaving sun-induced fluorescence? Remote Sens. Environ. 166, 8–21. http://dx.doi.org/10.1016/j.rse.2015.06.002, URL https://www.sciencedirect.com/science/article/pii/S0034425715300328.

Vilfan, N., van der Tol, C., Muller, O., Rascher, U., Verhoef, W., 2016. Fluspect-B: A model for leaf fluorescence, reflectance and transmittance spectra. Remote Sens. Environ. 186, 596–615. http://dx.doi.org/10.1016/j.rse.2016.09.017.

Yang, P., Prikaziuk, E., Verhoef, W., Van der Tol, C., 2021a. SCOPE 2.0: a model to simulate vegetated land surface fluxes and satellite signals. Geosci. Model Dev. 14 (7), 4697–4712. http://dx.doi.org/10.5194/gmd-14-4697-2021.

Yang, P., Verhoef, W., Prikaziuk, E., van der Tol, C., 2021b. Improved retrieval of land surface biophysical variables from time series of Sentinel-3 OLCI TOA spectral observations by considering the temporal autocorrelation of surface and atmospheric properties. Remote Sens. Environ. 256, 112328. http://dx.doi.org/10.1016/j.rse.2021.112328.